# Luis Thomas

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## Education

**Ph.D. Astronomy** May 2022–Jun 2026
University Observatory Munich, Germany
Thesis: *RV follow-up of TESS planet candidates with the MaHPS spectrograph*
Advisor: Dr. Roberto Saglia
Grade: Summa Cum Laude (0.7)

**M.Sc. Planetary Science** Oct 2020–Oct 2021
University College London, UK
Thesis: *Detecting helium in exoplanet atmospheres with the Hubble Space Telescope*
Advisor: Dr. Angelos Tsiaras

**B.Sc. Physics** Oct 2017–Sep 2020
Ludwig-Maximilians University, Germany
Thesis: *Detecting exoplanets in the habitable zone around M-dwarfs*
Advisor: Dr. Roberto Saglia

## Publications

### Lead Author

5. **Thomas, L.**, Nielsen, L. D., Cridland, A., et al. (2026), *A&A* accepted
*Companion architectures of sub-Saturns: Distinct migration pathways across the Neptunian landscape*
4. **Thomas, L.**, Nielsen, L. D., Kellermann, H., et al. (2026), *A&A* 712 A157
*TOI-2147 b and TOI-6019 b: Two eccentric warm Jupiters detected and characterized with TESS and MaHPS*
3. **Thomas, L.**, Nielsen, L. D., Acuña-Aguirre, L., et al. (2025), *A&A* 703 A221
*Evidence of a gap in the envelope mass fraction of sub-Saturns*
2. **Thomas, L.**, Hébrard, G., Kellermann, H., et al. (2025), *A&A* 694 A143
*TOI-5108 b and TOI 5786 b: Two transiting sub-Saturns detected and characterized with TESS, MaHPS and SOPHIE*
1. **Thomas, L.**, Saglia, R., Pasquini, L., et al. (2024), *A&A* 686 A19
*Search for giant planets in M67 V: a warm Jupiter orbiting the turn-off star S1429*

### Co-Author

5. Almenara, J., Castro-González, A., Mardling, R., et al. (incl. **Thomas, L.**) (2026), *A&A* 714 A190
*HD 148797: A bright F-type star with two moderate-period low-density sub-Jovian planets*
4. Grouffal, S., Santerne, A., Bourrier, V., et al. (incl. **Thomas, L.**) (2025), *A&A* 701 A173, 2025
*The star HIP 41378 potentially misaligned with its cohort of long-period planets*
3. Ehrhardt, J., **Thomas, L.**, Kellermann, H., et al. (2024), *A&A* 692 A220
*Confirmation of four hot Jupiters detected by TESS using follow-up spectroscopy from MaHPS at Wendelstein together with NEID and TRES*
2. Korth, J., Chaturvedi, P., Parviainen, H., et al. (incl. **Thomas, L.**) (2024), *ApJL* 971 L28
*TOI-1408: Discovery and Photodynamical Modeling of a Small Inner Companion to a Hot Jupiter Revealed by Transit Timing Variations*
1. Edwards, B., Changeat, Q., Tsiaras, A., et al. (incl. **Thomas, L.**) (2023), *AJ* 166 158
*Characterising a World Within the Hot Neptune Desert: Transit Observations of LTT 9779 b with HST WFC3*

# Teaching & Mentorship

## Teaching

Basic Lab Exoplanet Detection | Supervisor 2024 – 2026
Basic Lab Exoplanet Detection | Redesign and update of the lab manual May 2025

## Student Mentoring

Paul Lotz | BSc Thesis Supervisor 2026
Mael Minig | BSc Thesis Supervisor 2026
Jebraan Mudholkar | MSc Thesis Co-supervisor since 2025
Yutong Liu | MSc Thesis Co-supervisor 2025 – 2026
Juliana Ehrhardt | MSc Thesis Supervisor 2022 – 2024
Laura Schöller | MSc Thesis Supervisor 2023 – 2024
Nicolas Dowling | MSc Thesis Co-supervisor 2022 – 2023
Khushi Marlecha | MSc Thesis Co-supervisor 2022 – 2023

# Selected Talks & Posters

**The Formation, Life and Fate of Exo-Neptunian Systems (Workshop)**, Leiden, NL Sep 2026
*Companion architectures of sub-Saturns;* **contributed talk**.

**Multiple Planet Systems (Conference)**, Sofia, Bulgaria Sep 2026
*Planet Multiplicity Across the Neptune Desert, Ridge, and Savanna;* **contributed talk**.

**Exocoffee (Journal Club)**, MPIA, Heidelberg, Germany Jul 2026
*Evidence of a gap in the envelope mass fraction of sub-Saturns;* **invited seminar**.

**Astrophysikalisches Kolloquium (Colloquium)**, Georg-August-Universität, Göttingen, Germany Jul 2026
*The Sub-Saturn Population: Interior Structure and System Architecture as Probes of Planet Formation;* **invited seminar**.

**ABC forum (Seminar)**, Astrobiology Center, Tokyo, Japan May 2026
*The Sub-Saturn Population: Interior Structure and System Architecture as Probes of Planet Formation;* **invited seminar**.

**Star-Planet Exchange (SPEX) (Seminar)**, ESO Garching, Germany Dec 2025
*Exploring the formation of warm sub-Saturns through interior structure modeling;* **invited seminar**.

**The Fate of Neptunian Worlds (Conference)**, Warwick, UK Sep 2025
*Exploring the formation of warm sub-Saturns through interior structure modeling;* **contributed talk**.

**EPRV 6 (Conference)**, Porto, Portugal Jun 2025
*Exoplanet Detection and mass measurement with the MaHPS spectrograph at Wendelstein;* **poster**.

**Austrian early career conference (Conference)**, Salzburg, Austria Mar 2024
*Radial Velocity Measurements with the MaHPS spectrograph on the Wendelstein Observatory;* **contributed talk**.

**Exoplanets by the Lake I (Summer school)**, Munich, Germany Aug 2023
*Radial Velocity Measurements with the MaHPS spectrograph on the Wendelstein Observatory;* **contributed talk**.

# Observing & Awarded telescope time

**PI** • HARPS • 9.5 hrs Populating the runaway accretion gap (Prog. 117.2A28)
**Co-I** • ESPRESSO • 122.5 hrs Probing Planetary Evolution: Masses of Young Transiting Exoplanets with ESPRESSO (Prog. 117.2AJ3)
**Co-I** • ESPRESSO • 28 hrs Planet evolution in- and around the desert: measuring masses of the young Neptunes orbiting TOI-942 (Prog. 114.277K)

**Wendelstein:** 52 nights as the sole on-site observer at Wendelstein Observatory. Responsible for the execution of science programs on the 2.1 m and 43 cm telescopes as well as telescope operating and instrument maintenance.

## Service & Outreach

**Scientific Referee**
Astronomy and Astrophysics

**Guided Tours at the Wendelstein Observatory**
Giving guided tours to the general public of the observatory at Mount Wendelstein. Explaining the science programs as well as the available telescopes and instruments.

**Techmax 40: Exoplaneten**
Contribution to the Techmax issue on exoplanets, a booklet by the Max-Planck Society to promote science topics.